events will cause at least half the damages and losses.

From Reality to Representation[*]

Let me take another angle. The passage from theory to the real world presents two distinct difficulties: inverse problems and pre-asymptotics.

Inverse Problems. Recall how much more difficult it is to re-create an ice cube from the results of the puddle (reverse engineering) than to forecast the shape of the puddle. In fact, the solution is not unique: the ice cube can be of very many shapes. I have discovered that the Soviet-Harvard method of viewing the world (as opposed to the Fat Tony style) makes us commit the error of confusing the two arrows (from ice cube to puddle; from puddle to ice cube). It is another manifestation of the error of Platonicity, of thinking that the Platonic form you have in your mind is the one you are observing outside the window. We see a lot of evidence of confusion of the two arrows in the history of medicine, the rationalistic medicine based on Aristotelian teleology, which I discussed earlier. This confusion is based on the following rationale. We assume that we know the logic behind an organ, what it was made to do, and thus that we can use this logic in our treatment of the patient. It has been very hard in medicine to shed our theories of the human body. Likewise, it is easy to construct a theory in your mind, or pick it up from Harvard, then go project it on the world. Then things are very simple.

This problem of confusion of the two arrows is very severe with probability, particularly with small probabilities.[*]

As we showed with the undecidability theorem and the self-reference argument, in real life we do not observe probability distributions. We just observe events. So I can rephrase the results as follows: we do not know the statistical properties—until, of course, after the fact. Given a set of observations, plenty of statistical distributions can correspond to the exact same realizations—each would extrapolate differently outside the set of events from which it was derived. The inverse problem is more acute when more theories, more distributions can fit a set of data, particularly in the presence of nonlinearities or nonparsimonious distributions.[†] Under nonlinearities, the families of possible models/parametrization explode in numbers.[‡]

But the problem gets more interesting in some domains. Recall the Casanova problem in Chapter 8. For environments that tend to produce negative Black Swans, but no positive Black Swans (these environments are called negatively skewed), the problem of small probabilities is